10 The Turnover Room

Alpha, beta & gamma diversity

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 35 min	**+ 28 min**	The Diversity Room (09), which measures one community the same way.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/population-laboratories/` — then the Turnover tab

Several sites, each with its own community. γ is the diversity of all of them pooled, α that of a typical site, and $\beta = \gamma/\alpha$ how many completely distinct sites they amount to. This guide asks how a landscape's diversity is built from its parts, and what happens when the sites drift apart.

LEARNING OBJECTIVES

After the core path you should be able to

- Define α, β and γ diversity and use $\gamma = \alpha \times \beta$.
- Show that local diversity does not predict regional diversity.
- Explain how the choice of measure (richness, Shannon, Simpson) changes β.
- Show that neutral drift in isolated sites creates turnover, and that immigration from a shared pool removes it.

The extension adds

- Calculate α, β and γ for a small landscape by hand.
- Connect β to F_{ST} in the Migration Room.
- Discuss the several-small-or-single-large question, and what the model leaves out of it.

SUGGESTED ROUTES

- **40-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Conservation planning seminar:** step 4 with questions 3 and 6.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 Identical sites and distinct sites

10 min

SET UP

Step 1 of the room, $N = 4$ sites, measure by Shannon. Preset **Identical**, then **Complete turnover**.

PREDICT FIRST

In each landscape, how will γ compare with α? What will β be?

RUN

Choose each preset and read the three numbers.

LOOK AT

Diversity at each scale: the site bars, α and γ.

EXPLAIN

Why does adding identical sites add no diversity to the landscape?

2 The measure matters

8 min

SET UP

Preset **Nested**, then **Gradient**. For each, switch the measure between Richness, Shannon and Simpson.

PREDICT FIRST

Nested sites differ only in their rare species. By which measure will their β be largest? And for the gradient, where each site has its own commonest species?

RUN

Read β under each measure for both presets.

EXPLAIN

What does the direction of the difference tell you about *which* species turn over?

3 Drift makes turnover

10 min

SET UP

Step 2 of the room. Each site starts from **The pool**, J = 200, immigration m = 0, 300 generations.

PREDICT FIRST

Every site starts with the same community. What will happen to α and to β as the sites drift?

RUN

Press **Run**.

LOOK AT

Each site now and *β over time*.

EXPLAIN

Why do sites that start alike lose *different* species?

4 Several small or single large?

7 min

SET UP

The run you just made.

PREDICT FIRST

The large site holds as many individuals as all the small sites together. After 300 generations, which will hold more species: the small sites together (γ) or the large site?

RUN

Read *Several small or single large?* and the middle panel's two lines through time.

EXPLAIN

Under pure chance, which wins in the end, and why? What real risks to small sites does the model leave out?

EXTENSION · MSC

5 By hand

10 min

SET UP

Build two sites with N = 2: site 1 holds species 1 and 2 equally; site 2 holds species 2 and 3 equally.

PREDICT FIRST

Calculate γ, α and β by richness, by Shannon and by Simpson.

RUN

Drag the bars to make this landscape and read the numbers.

EXPLAIN

Why do the three measures give different β for the same landscape?

6 Immigration erases turnover

10 min

SET UP

Step 2. Each site starts from **Its own row** with the Gradient preset; immigration m = **0.01**.

PREDICT FIRST

Each site starts different. What will a shared stream of immigrants do to β?

RUN

Run, and then again with $m = 0$.

EXPLAIN

“Mixing makes places locally richer and regionally the same.” Explain, with your two runs.

7 β and F_{ST}

8 min

SET UP

Step 2 from the pool, $m = 0$, 0.002 and 0.02.

PREDICT FIRST

How will β at the end change with m?

RUN

Run each.

EXPLAIN

Compare with the Migration Room, where F_{ST} falls as migration rises. Why do the two measure the same thing at different levels of organisation?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Two regions each have an average of 10 species per site. Can one region have far more species overall? Explain with β.
2. BSc Why is β in this room counted in “sites” rather than in species?
3. BSc A conservation agency must choose between one large reserve and four small ones of the same total area. What does this room say, and what does it leave out?
4. MSc Additive partitions ($\beta = \gamma - \alpha$) and pairwise indices (Jaccard, Sørensen) also measure turnover. Why can’t values from different conventions be compared?
5. MSc Sampling inflates β. Explain why, using the Diversity Room’s lesson about richness.
6. MSc Real turnover is mostly driven by environmental differences rather than drift. How would you distinguish the two in field data?

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** Identical: $\gamma = \alpha$, $\beta = 1$. Complete turnover: $\gamma = 4\alpha$, $\beta = N = 4$.
- **Step 2.** Nested: β largest by richness (the sites differ only in rare species, which Shannon and Simpson discount). Gradient: largest by Simpson (each site's commonest species is its own).
- **Step 3.** α falls as each site loses species; β climbs because the sites lose different species by chance.
- **Step 4.** The large site often leads early, but with no immigration the small sites together end up holding more: each ends with a species of its own, the large site with one. Edge effects, area-demanding species and single disasters are all missing.
- **Step 5.** Richness: $\gamma = 3$, $\alpha = 2$, $\beta = 1.5$. Shannon: $\gamma \approx 2.83$, $\alpha = 2$, $\beta \approx 1.41$. Simpson: $\gamma \approx 2.67$, $\alpha = 2$, $\beta \approx 1.33$.
- **Step 6.** Immigration from the pooled landscape pulls every site towards the same community, so β falls towards 1.
- **Step 7.** β falls as m rises, just as F_{ST} falls with migration: both measure how different places are, for species and for alleles.